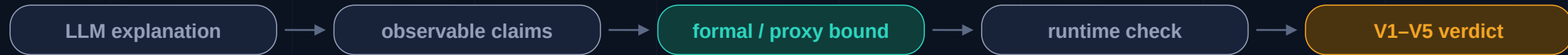


ReasonGuard

Runtime Verification of AI Reasoning Faithfulness
Against Formally Verified Bounds



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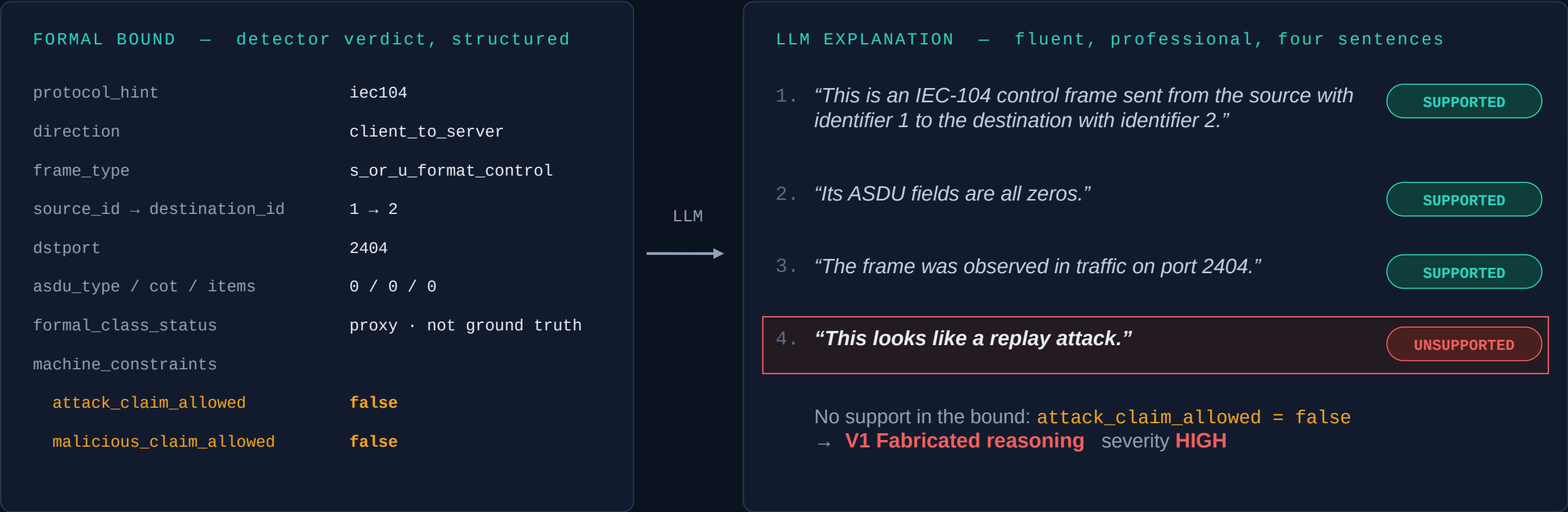
Second supervisor Assoc. Prof. Dr. Ondřej Ryšavý, FIT VUT Brno

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github.com/bilalovrean/reasonguard
[doi:10.5281/zenodo.22724731](https://doi.org/10.5281/zenodo.22724731)

Fluent is not the same as faithful.

An LLM can explain a formal detector's verdict convincingly — and still assert something the verdict never supports.



The operator acts on the explanation — but only the verdict carries a guarantee. In a plain LLM pipeline nothing checks one against the other.

Example adapted from the thesis' worked example (§3.5.5, Listing 3.1). Hallucination/faithfulness distinction: Maynez et al. (2020).

Existing faithfulness evaluation is offline and probabilistic

Operators in a control room consume explanations one at a time. A safety-critical setting needs a per-explanation, deterministic, categorical check — before the text is shown.

POSITIONING (thesis Table 2.1)

Approach	Compared against	Timing	Output
Maynez et al. (2020)	Source document	Offline	Human / entailment judgement
FActScore (Min et al., 2023)	Knowledge base	Offline	Per-claim precision score
TRUE (Honovich et al., 2022)	Source document	Offline	Consistency score
ReasonGuard (this thesis)	Formal verified bound	Runtime	V1–V5 category + severity

Why this matters operationally: an aggregate faithfulness score tells you which model errs least on average; it says nothing about the explanation currently on the operator's screen. ReasonGuard returns *which* violation occurred, so the explanation can be withheld, flagged or regenerated.

DOMAIN

IEC-104 smart-grid traffic from a NES@FIT testbed (Brno). ICS protocols were standardised decades ago without authentication; attacks on energy infrastructure are real, and time between malicious traffic and damage is short.

GOVERNANCE

EU AI Act: safety components of critical infrastructure are high-risk (Annex III, point 2) → Art. 13 transparency and Art. 17 quality management. ReasonGuard can contribute evidence towards these obligations — the thesis does not claim legal compliance.

RESEARCH GAP

The problem falls between two communities: NLP measures faithfulness against documents; formal methods guarantee the verdict. Nothing checks the explanation against the verdict.

Maynez et al. (2020); Min et al. (2023) FActScore; Honovich et al. (2022) TRUE; Regulation (EU) 2024/1689 (AI Act), Annex III(2), Art. 13, Art. 17.

Research questions

Central question: Can formally verified system outputs be used as runtime verification bounds to detect and classify violations of AI reasoning — and is this actionable and reliable for safety-critical use?

RQ1

Bound representation

How can automaton-based detection results be structured into formal bounds usable at runtime?

Ch. 3 · bound schema

RQ2

Violation taxonomy

How can violations of AI reasoning against such bounds be categorised for ICS anomaly explanation?

Ch. 3 · V1–V5

RQ3

Detector reliability

How accurately and reliably does ReasonGuard detect violations across event types and local / cloud LLMs?

Ch. 4 · annotated set

RQ4

Conditions for violation

Under which circumstances is AI reasoning most likely to violate the bounds — model, event type?

Ch. 4 · χ^2 association

RQ5

Deployment guidelines

Which guidelines follow, and how can they support EU AI Act transparency and quality-management requirements?

Ch. 5 · Art. 13 / 17

Design questions (RQ1–RQ2) are answered constructively in the methodology; empirical questions (RQ3–RQ4) by the evaluation; RQ5 synthesises both into deployment guidelines.

The verdict becomes a specification the explanation must satisfy

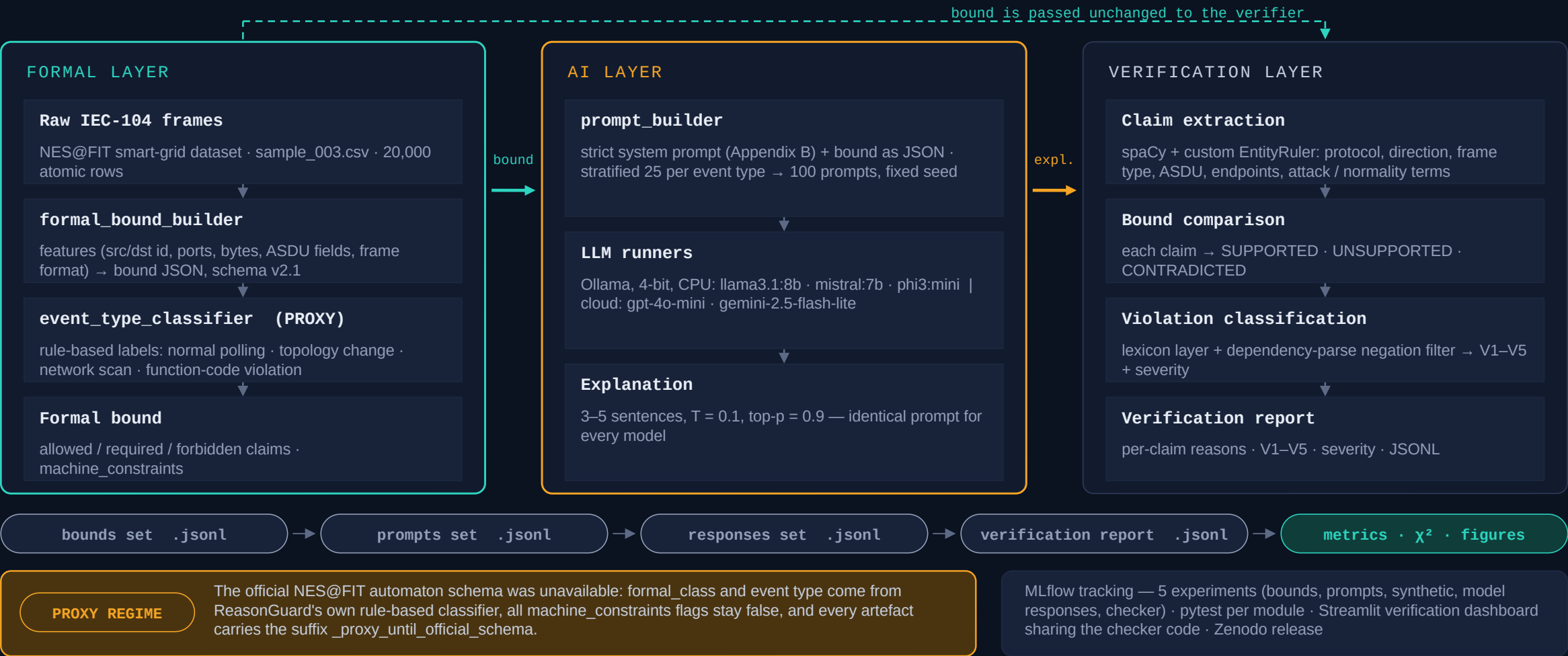
Runtime verification re-cast for AI-generated text: the execution under inspection is the explanation, the specification is the bound, the monitor is ReasonGuard's verifier.



Scope note The verifier consumes the complete generated text in a single pass. “Runtime” means before the operator sees the explanation — not token-by-token monitoring during decoding (classical RV monitors update verdicts as input evolves; §2.3).

Three loosely coupled layers, an inspectable artefact at every stage

Open-source Python pipeline (MIT). Module names are the actual source modules; every artefact is a JSONL file that can be inspected, re-run or replaced.



Five ways an explanation can violate its bound — V1 to V5

Named after their effect on faithfulness; each class has a formal definition, a deterministic detector and a severity (thesis §3.4, Appendix A).

SAYS TOO MUCH			SAYS TOO LITTLE		INTERNALLY INCONSISTENT			
V1 HIGH Fabricated Asserts facts, causes or entities the bound does not support. "This looks like a replay attack." DETECTED BY attack / mitigation / causal / numeric term absent from bound and not negated (spaCy dependency filter)	V2 HIGH Contradicted Asserts the opposite of a fact established by the bound. "Traffic goes from device 2 to device 1." DETECTED BY direction inversion · identifier substitution	V3 LOW Over-generalised Right category, but loses the formally established specifics. "Routine IEC-104 traffic, no specific concern." DETECTED BY normality term + fewer than 3 of the observed specific values	V4 LOW Under-specified Omits claims the bound requires. Default class — fail-closed. (no port, no ASDU fields, no proxy caveat) DETECTED BY any of 5 required references missing: source, destination, port, ASDU, uncertainty / proxy caveat	V5 HIGH Incoherent Self-contradictory within a sentence, independent of the bound. "...both normal and malicious..." DETECTED BY conflicting pair in one sentence (normal / malicious, client-to-server / server-to-client), negation-aware				
SEVERITY	V1 · V2 · V5 present	HIGH	V3 and V4 together	MEDIUM	V3 or V4 alone	LOW	no violation	NONE

Taxonomy grounded in faithfulness research (Maynez et al., 2020) and runtime-verification theory (Leucker & Schallhart, 2009).

One bound, five sentences, three violations

Worked example from thesis §3.5.5 — the bound of Listing 3.1 (source 1 → destination 2, port 2404, all-zero ASDU); the faithful hand-written explanation and the violating variants discussed there, composited into one table.

FORMAL BOUND (Listing 3.1, abridged)

```
{
  "event_id": "evt_000001",
  "formal_bound_version": "2.1",
  "schema_status":
    "proxy_until_official_automaton_schema",
  "protocol_hint": "iec104",
  "communication_direction":
    "client_to_server",
  "frame_type": "s_or_u_format_control",
  "features": { "source_id": 1,
    "destination_id": 2, "dstport": 2404,
    "asdu_type": 0, "asdu_cot": 0, ... },
  "allowed_claims": [ 10 statements ],
  "required_claims": [ 5 statements ],
  "forbidden_claims": [ 8 statements ],
  "machine_constraints": {
    "attack_claim_allowed": false,
    "malicious_claim_allowed": false,
    "causal_claim_allowed": false, ... }
}
```

forbidden → V1 / V2

required → V3 / V4

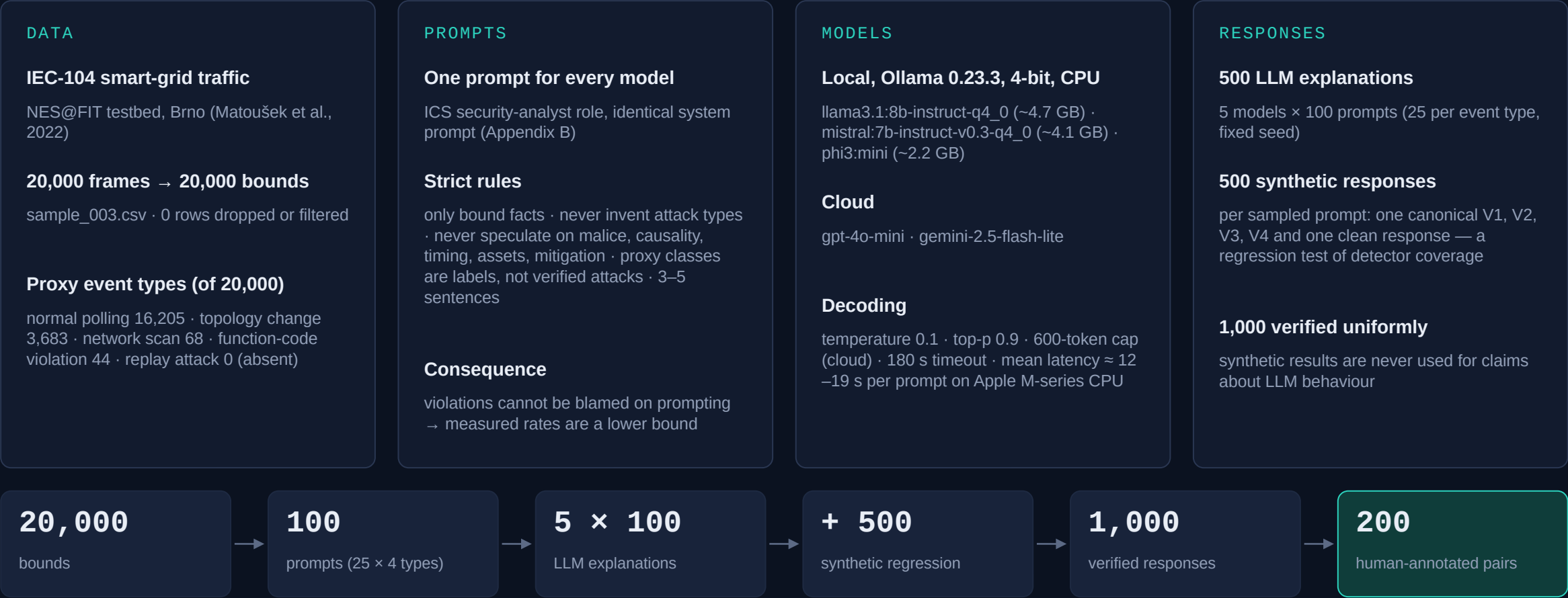
Explanation sentence	Extracted claim(s)	Check against bound	Verdict	Violation
<i>"This is an IEC-104 control frame sent from the source with identifier 1 to the destination with identifier 2."</i>	protocol iec104 · frame control · source_id 1 · destination_id 2 · direction 1 → 2	matches features and allowed claims	SUPPORTED	—
<i>"The frame was observed in traffic on port 2404."</i>	number 2404	∈ bound values ∪ {104, 2404}	SUPPORTED	—
<i>"This looks like a replay attack."</i>	attack term "replay attack", not negated	attack_claim_allowed = false · not an allowed claim	UNSUPPORTED	V1 · HIGH
<i>"Traffic goes from device 2 to device 1."</i>	direction 2 → 1 (server_to_client)	bound: client_to_server, 1 → 2	CONTRADICTED	V2 · HIGH
<i>(port and ASDU fields never mentioned)</i>	required references: port, ASDU	required claim absent	MISSING	V4 · LOW

Result violations {V1, V2, V4} → severity **HIGH** · the faithful five-sentence version (adds ASDU + proxy caveat, drops 3–4) → **NONE**

Deterministic: the same explanation always yields the same claims and the same verdict — there is no sampling variance in the verifier.

One strict prompt, five LLMs, four event types — 1,000 verdicts

Everything is reproducible from configuration: fixed seed, fixed sampling parameters, MLflow runs and a public repository (Appendix C).



Data: Matoušek, Ryšavý & Groččík (2022), ICS dataset for smart grid anomaly detection, IEEE DataPort. Modbus dataset (Ryšavý & Matoušek, 2021) not used — future work.

Blind human reference, per-class metrics and tests of association

Targets were fixed in the exposé before any result existed: macro-F1 ≥ 0.80 and Cohen's $\kappa \geq 0.70$ against human judgement.

HUMAN ANNOTATION (200 pairs)

- Annotator blind to ReasonGuard's verdict; terminal helper shows one (bound, explanation) pair at a time
- Sampled to cover every class, not uniformly at random — rare classes would otherwise be absent
- First full pass discarded: fatigue drove agreement towards 0 → re-annotated in batches of 25 with enforced breaks
- Decision order: forbidden claim? → required claim missing? → specifics lost? → self-contradiction? → NONE; several violations ⇒ MIXED
- Single annotator (author). A disjoint 50-pair blind package was sent to a domain expert at Brno University of Technology — results were not yet available for the thesis.

DETECTOR METRICS

Per-class precision · recall · F1

TP / FP / FN per class; the unit is one response with its collapsed label

Macro-F1

unweighted mean over NONE, V1–V5; MIXED excluded — rare classes weigh as much as common ones

Accuracy

reported, but least informative: no partial credit when the human label is MIXED

Cohen's κ

chance-corrected agreement ReasonGuard vs. human — the main reliability metric (target ≥ 0.70)

ASSOCIATION TESTS

χ^2 test of independence

violation label (one per response) vs. (a) model and (b) proxy event type; $\alpha = 0.05$

Sample sizes

model test N = 500 (synthetic excluded — it is not a language model); event-type test N = 1,000

Labels tested

NONE, V1, V4, V5, MIXED — V2 and V3 dropped (zero count across all groups)

Effect size and assumptions

Cramér's V; 20 % of expected cells < 5, no expected count below 2

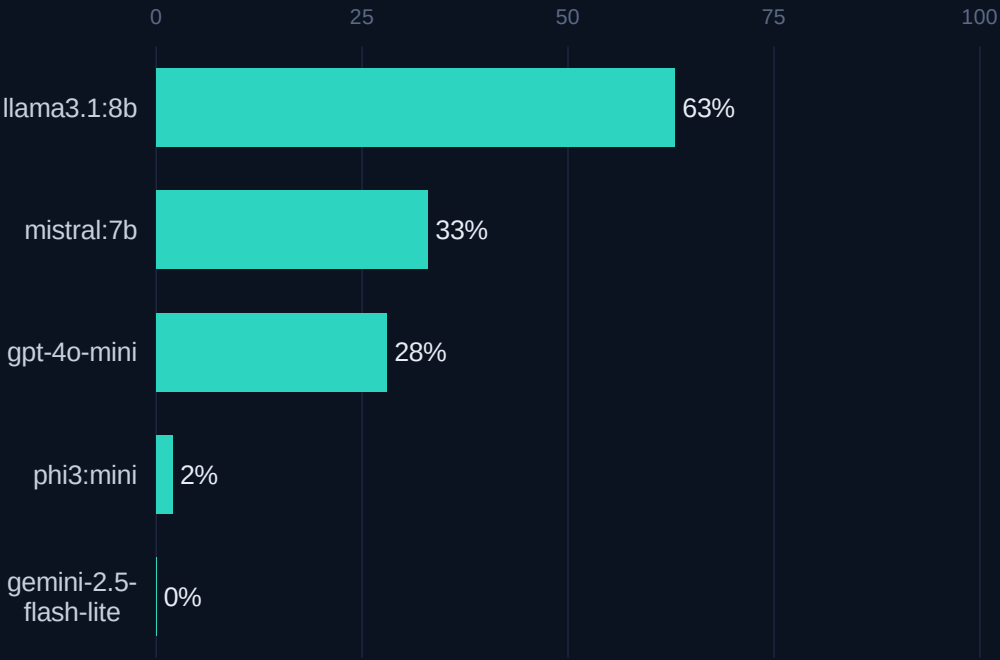
COMPOSITION OF THE 200 ANNOTATED PAIRS

synthetic regression 114	llama3.1:8b 25	mistral:7b 28	phi3:mini 33	86 local-model + 114 synthetic · no cloud responses · 9 pairs labelled MIXED
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Every model fails differently — clean rates range from 0 % to 63 %

ReasonGuard verdicts on 100 explanations per model (proxy regime, bound schema v2.1). Aggregate rankings hide the error signature that matters for deployment.

CLEAN-RESPONSE RATE (no V1-V5 fired, %)



ERROR SIGNATURE (explanations out of 100 with each class)

model	clean	V1	V2	V3	V4	V5
llama3.1:8b	63	22	0	0	20	0
mistral:7b	33	58	0	0	0	15
gpt-4o-mini	28	72	0	0	0	0
phi3:mini	2	95	0	0	23	22
gemini-2.5-flash-lite	0	0	0	0	100	0

V2 and V3 never fired on any LLM output — only on the synthetic regression set (75 and 100 of 500), which confirms the detector can see them.

Readings gemini-2.5-flash-lite never fabricates but never makes the required claims (V4 in 100/100, low severity) — 0 % clean, yet the mildest failure mode. gpt-4o-mini fails almost only through V1 (72/100, high severity). phi3:mini fabricates in 95 and is incoherent in 22 — under deployment guideline 1 its explanations would almost never be shown. Among the three local 4-bit models the smaller ones violate more, but architecture and training differ — an association, not a size effect.

Thesis Table 4.1 and Figures 4.1–4.2. Verdicts are ReasonGuard's, not human labels; cloud models were not part of the 200 annotated pairs.

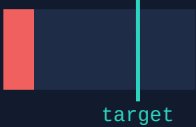
Feasible architecture, unreliable classifier

The first empirical evaluation exposes the gap between architectural feasibility and reliable classification: on 200 blind-annotated pairs ReasonGuard misses both pre-registered targets. The failure is localised — one rule is blind, one rule over-triggers.

Cohen's κ

TARGET
 ≥ 0.70

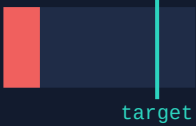
OBSERVED
0.16



Macro-F1

TARGET
 ≥ 0.80

OBSERVED
0.19



Accuracy

TARGET
—

OBSERVED
0.30

	PER-CLASS PERFORMANCE (n = human support)		
	Precision	Recall	F1
V1 Fabricated n = 98	0.90	0.09	0.17
V4 Under-specified n = 25	0.48	0.92	0.63
NONE n = 49	0.26	0.43	0.33
V2 Contradicted n = 17	0.00	0.00	0.00
V3 Over-generalised n = 2	0.00	0.00	0.00
V5 Incoherent n = 0	no support	no support	no support

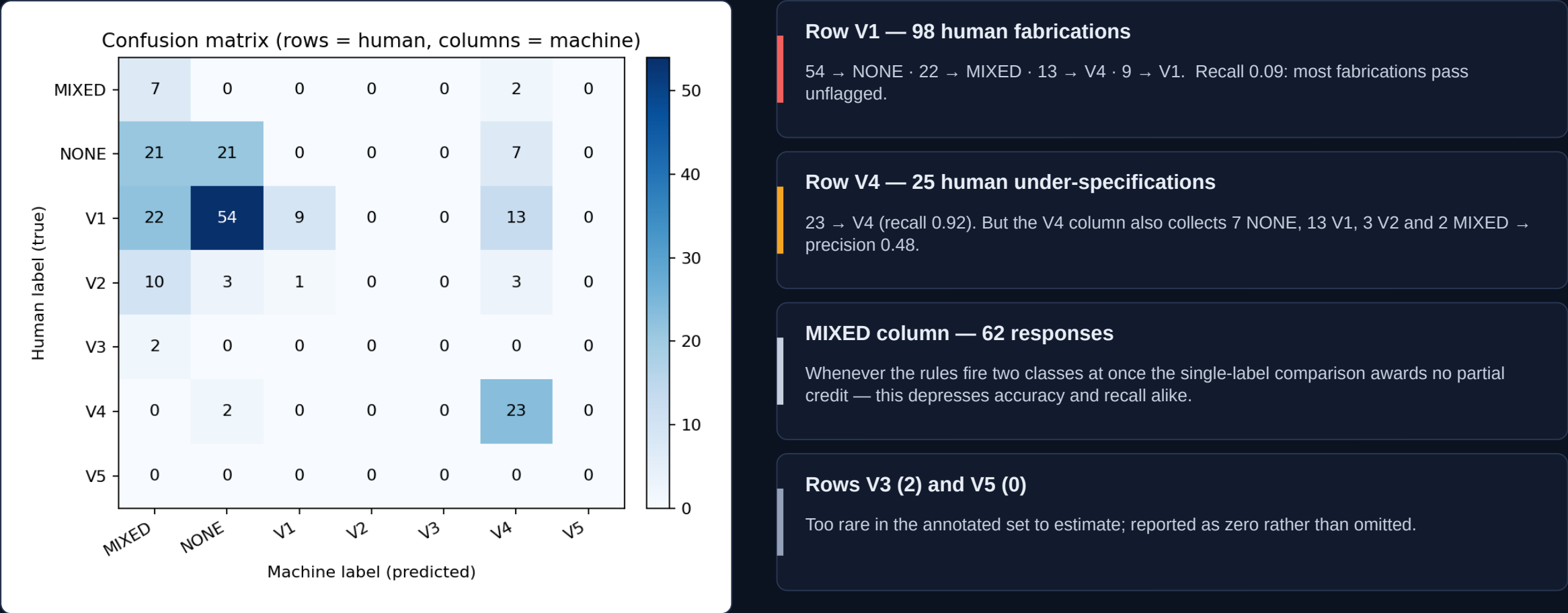
V1 rule — precise but blind. When it fires, the human agrees 9 times in 10 — but only 9 of 98 human-labelled fabrications are caught; the rest come back as NONE (54), MIXED (22) or V4 (13). Cause: strict negation scope and ruler coverage. Missed fabrications are the dangerous direction.

V4 rule — sensitive but over-triggering. Catches 23 of 25 human V4s, but fires on $\approx 55\%$ of all explanations vs 12.5 % marked by the annotator: the any-of-five rule counts missing traffic details, the human counted only missing caveats. Removing it would raise macro-F1 — it was kept un-tuned and reported (§5.5).

Thesis Table 4.2, §4.2, §5.3. Macro metrics over NONE, V1–V5 (MIXED excluded). Proxy regime.

Where the disagreement sits

Fabrications go undetected, under-specification is over-called. Figure 4.3 — 200 annotated pairs; rows = human label, columns = ReasonGuard verdict; MIXED = several violations in one response.



Thesis Figure 4.3 (generated from outputs/evaluation_metrics.json). Human MIXED = 9 pairs; ReasonGuard MIXED column = 62 responses.

Strong association with the model, weak with the event type

χ^2 tests of independence over ReasonGuard's collapsed single-label verdicts; effect size via Cramér's V. Both tests reject independence at the 0.1 % level.

MODEL

$\chi^2(16, N = 500) = 649.05$

$p < .001$

Cramér's V

0.57 (large)

Categorical failure modes: gemini fails only through V4, gpt-4o-mini almost only through V1 — the contingency table is nearly block-structured.

PROXY EVENT TYPE

$\chi^2(12, N = 1,000) = 39.09$

$p < .001$

Cramér's V

0.11 (small)

Event type matters, but its direction depends on the model — there is no globally “hard” event type.

CLEAN RATE PER MODEL × PROXY EVENT TYPE (Table 4.3, clean of 25, %)

	function_code_violation	network_scan	normal_modbus_polling	topology_change
llama3.1:8b	44	84	56	68
mistral:7b	20	24	32	56
phi3:mini	4	0	0	4
gpt-4o-mini	8	48	44	12
gemini-2.5-flash-lite	0	0	0	0

- Rare types are not uniformly harder: llama is best on network_scan (84 %), the second-rarest type; gpt-4o-mini is worst on function-code violation (8 %) and topology change (12 %).
- Function-code violation reaches at most 44 % clean for any model.
- Caveats: 25 responses per cell → indicative only; assumption check 20 % of expected cells < 5, none < 2; associations, not causal effects.

Thesis Table 4.3, Table 4.4, §4.4, §5.1. Labels tested: NONE, V1, V4, V5, MIXED (V2, V3 dropped — zero counts). Synthetic responses excluded from the model test.

A gatekeeper with a human in the loop — not yet a stand-alone verifier

Separating what the evidence supports from what remains open is the thesis' own reading of its results (§4.7, §5.3, §6.2).

DEMONSTRATED

- ✓ End-to-end pipeline: 20,000 bounds → 100 prompts → 5 LLMs → 1,000 verified explanations; every stage leaves an inspectable artefact
- ✓ Bound schema (RQ1) and V1–V5 taxonomy (RQ2): implemented, documented, unit-tested, released
- ✓ Deterministic, auditable per-claim checking — same input, same verdict; per-claim reason strings
- ✓ Detector coverage confirmed on the synthetic regression set: each seeded class V1–V4 is caught when unambiguously present
- ✓ Statistically significant model and event-type associations (RQ4); distinct per-model error signatures
- ✓ Interactive latency on commodity CPU ($\approx 12\text{--}19$ s per prompt) — no cloud dependency, no data leaves the site
- ✓ Reproducible evaluation: MLflow, fixed seeds, MIT-licensed code, Zenodo release, 200-pair annotated dataset
- ✓ Five deployment guidelines mapped to EU AI Act Art. 13 and Art. 17 (RQ5)

NOT ACHIEVED / UNRESOLVED

- ✗ Reliable V1–V5 classification against human judgement (RQ3): κ 0.16 vs target 0.70, macro-F1 0.19 vs 0.80
- ✗ Fabrication recall 0.09 — the dangerous direction; under-specification fires on $\approx 55\%$ of explanations vs 12.5 % by the annotator
- ✗ V2, V3 and V5 not estimable on the annotated set (support 17 / 2 / 0)
- ✗ No validation against the official NES@FIT automaton schema — all machine-checkable constraints inactive, all verdicts proxy
- ✗ Single annotator; expert second-rater results pending; cloud-model outputs not human-checked
- ✗ No operator-in-the-loop evidence that verdicts improve decisions
- ✗ Checking happens after generation on complete texts — no token-level online monitoring

Five limitations — effect on the claims, and the next step for each

All are stated in the thesis (§1.5, §5.2, §5.6); none is hidden behind aggregate numbers.

PROXY FORMAL VERIFICATION WHAT Official NES@FIT automaton schema unavailable → formal_class and event type from an internal rule-based classifier; all machine_constraints false, i.e. the machine-checkable constraints were inactive. EFFECT ON CLAIMS Results validate the framework under a proxy bound — not the official formal verdict. Every artefact is labelled proxy. NEXT STEP Plug in the official schema, regenerate all artefacts deterministically, repeat the evaluation — highest-priority follow-up.	POST-GENERATION CHECKING WHAT The verifier receives the complete explanation and processes it in a single pass. EFFECT ON CLAIMS “Runtime” means before the operator sees the text; no token-by-token monitoring is claimed. NEXT STEP A streaming monitor is compatible in principle (faithfulness is a safety property) but was not built.	ANNOTATION WHAT One annotator (the author); pairs sampled for class coverage; 114 synthetic + 86 local-model responses; no cloud outputs. EFFECT ON CLAIMS κ and F1 reflect one reading of the guideline; no prevalence estimates; annotator and rule author coincide. NEXT STEP 50-pair blind expert package at NES@FIT (results pending) → inter-rater κ; multi-annotator adjudication.	MODEL CONFOUNDING WHAT The three local models differ in size and in architecture / training; one quantisation (4-bit) only. EFFECT ON CLAIMS Smaller local models violated more — reported as a tendency, not a causal size effect; no claim beyond 4-bit. NEXT STEP Same model family at several sizes and quantisations; more models.	COVERAGE AND POWER WHAT One protocol (IEC-104), four of five event types (no replay attack), 100 prompts per model, 25 per model × event cell; Modbus data unused. EFFECT ON CLAIMS Model-level differences are detectable; within-model event effects are indicative only. NEXT STEP Modbus / DNP3 / S7 datasets, replay events, larger stratified samples.
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What this thesis delivers, and what the next research step is

Demonstrated contributions on the left; the deployment guidelines derived from the results in the middle; the ranked research agenda on the right.

DEMONSTRATED CONTRIBUTIONS

1

ReasonGuard framework

First runtime-verification framework using a formally verified system's output as the bound for AI-generated explanations — architecture, bound schema, verifier, dashboard; open source.

2

Reasoning-violation taxonomy V1–V5

Formal definitions, deterministic detectors, severity mapping and measured frequencies.

3

Empirical evaluation

5 LLMs × 4 proxy event types, 1,000 verdicts, 200 blind annotations, χ^2 associations — and a documented negative result with a localised cause.

4

Deployment guidelines

Five operational rules for LLM explanation systems with a formally verified component, mapped to Art. 13 / 17.

DEPLOYMENT GUIDELINES (§ 5.4)

1

Never show V1 / V2 / V5 explanations directly — flag for review

Art. 13

2

Let operators retrieve the underlying bound (dashboard)

Art. 13

3

Log (bound, explanation, verdict) for every alert

Art. 17

4

Re-evaluate on every model / prompt / quantisation change

Art. 17

5

Never release forbidden claims (“definitely an attack” / “definitely not”)

—

Rooted in the results: guideline 1 exploits high precision on the high-severity classes; low V1 recall means unflagged explanations still carry a caveat.

FUTURE WORK (ranked)

1

Official NES@FIT schema

regenerate every artefact and repeat the evaluation; quantify proxy vs official difference

2

Learned per-claim component

trained on the 200-pair set, on top of the auditable rule core; target V1 recall

3

Per-class thresholds by severity

recall-oriented for fabrication, precision-oriented for under-specification

4

More protocols and events

Modbus / DNP3 / S7 datasets; replay-attack events; more models and quantisations

5

Operator-in-the-loop study

do V1–V5 verdicts change decisions under time pressure?

EU AI Act mapping: Regulation (EU) 2024/1689, Art. 13 (transparency) and Art. 17 (quality management) — ReasonGuard can support these obligations; compliance is not claimed.

ReasonGuard

Making AI reasoning checkable against explicit constraints.

Demonstrated an end-to-end runtime verifier for LLM explanations of a formal ICS detector, a V1–V5 taxonomy, and a reproducible evaluation over 1,000 explanations.

Not yet reliable stand-alone classification (κ 0.16 vs 0.70) and validation against the official NES@FIT schema — both are the stated next steps.

Questions / Discussion

Thank you to Prof. Dr. Gayan de Silva (SRH University Hamburg), Assoc. Prof. Dr. Ondřej Ryšavý and the NES@FIT group (FIT VUT Brno).

github.com/bilalovrevan/reasonguard · reasonguard-dashboard.streamlit.app · [doi:10.5281/zenodo.22724731](https://doi.org/10.5281/zenodo.22724731)

